

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2016-17

Paper No. BPT-101: Mechanics and Relativity

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer all questions. Symbols have their usual meanings.

Question 1. Answer any **four** of the following: [4 × 3.5 = 14]

- (a) What are fictitious (pseudo) forces and when do they come into play? Calculate the fictitious force and observed (total) force on a body of mass 5 kg in a frame of reference moving:
 - (i) vertically upwards with an acceleration of 4 m/s^2 .
 - (ii) vertically downwards with an acceleration of 4 m/s^2 .(Take $g = 9.8 \text{ m/s}^2$).
- (b) What is the rest mass of a photon? Give arguments in support of your answer.
- (c) What are central forces? Show that the angular momentum of a particle moving under the influence of a central force always remains constant.
- (d) In a nuclear reactor, hydrogen (in the form of water or heavy water) is used as a moderator. Give reasons.
- (e) Obtain an expression for the time period of a compound pendulum and discuss its advantages over a simple pendulum.
- (f) What are damped and undamped oscillations? Discuss with examples. In an oscillatory LCR circuit, $L = 0.2 \text{ H}$ and $C = 1.2 \mu\text{F}$. What is the maximum value of resistance R for the circuit to remain oscillatory?

Question 2.

- (a) What is a cantilever? A cantilever of rectangular cross-section is loaded at its free end. Obtain an expression for the depression of the cantilever at its free end. [9]
- (b) Calculate Poisson's ratio for silver. Given that Young's modulus for silver is $7.25 \times 10^{10} \text{ N/m}^2$ and its bulk modulus is $1.1 \times 10^{11} \text{ N/m}^2$. [5]

Question 3.

- (a) On the basis of the Lorentz transformation equations, discuss:
 - (i) length contraction.

(ii) time dilation.

[10]

- (b) A proton of rest mass 1.67×10^{-27} kg is moving with a velocity of $0.9c$. Find its relativistic mass and momentum. [4]

OR

- (a) What are the assumptions in Poiseuille's equation for the flow of a liquid through capillary tubes? Show that the coefficient of viscosity is given by:

$$\eta = \frac{\pi P r^4}{8 V L}$$

where V is the volume of the liquid flowing per unit time and the other symbols have their usual meanings. [8]

- (b) Explain how you will determine the coefficient of viscosity of water by Poiseuille's method in the laboratory. Discuss the effect of temperature on the viscosity of liquids. [6]

Question 4.

- (a) What do you mean by relativistic mass? Derive the relativistic mass-energy relation $E = mc^2$ and discuss its physical importance. [8]
- (b) If the relativistic kinetic energy is given by $K = mc^2 - m_0c^2$, show that its classical value (when $v \ll c$) reduces to $\frac{1}{2}m_0v^2$. [6]

OR

- (a) Derive an expression for the twisting couple per unit angular twist for a solid cylinder. Show that for cylinders of the same mass, material, and length, the twisting couple per unit twist is greater for a hollow cylinder than for a solid one. [8]
- (b) Obtain an expression for the horizontal deflection of a body falling freely under the action of gravity at a latitude λ due to the Coriolis force. [6]

Question 5.

- (a) Calculate the displacement suffered by a stone dropped from the top of a tower 100 m high due to the Coriolis force:
- (i) in latitude 60°N .
- (ii) at the equator.
- (Take angular velocity of the Earth $\omega = 7.29 \times 10^{-5}$ rad/s and $g = 9.8$ m/s²). [14]

OR

- (a) What do you understand by the laboratory (L) and centre of mass (C) frames of reference? Discuss two-dimensional elastic collisions in both frames of reference. Discuss

the advantages of studying collision processes in the centre of mass frame of reference.
[14]